27/7/24.

# Practical - 2

## Study of different types of network cables.

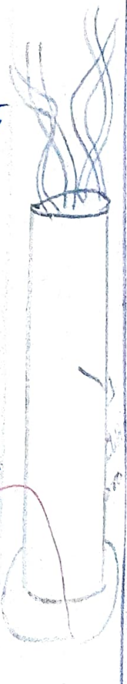
Aim:

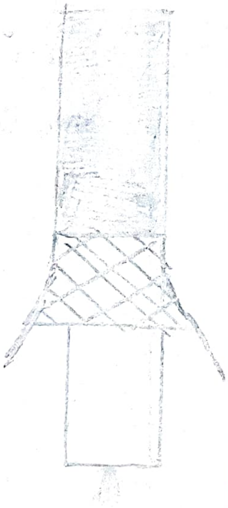
Study of different types of network cable

a) Understand different types of network cables

different types of cables used in networking are

1) Unshielded Twisted pair (UTP) cable.
2) Shielded Twisted pair (STP) cable
3) coaxial cable
4) Fibre optic cable.

| Cable type | Category | Maximum data Transmission | Adv/disadv | Appli use | Image |
|---|---|---|---|---|---|
| UTP | Category 3<br>Category 5<br>Category 5e | 10 Mbps<br>upto 100 Mbps | Adv:<br>* Cheaper in cost<br>* easy to install as they have a smaller overall diameter.<br>disadvantage<br>* More prone to EMI and Noise. | 10 Base-T Ethernet<br>Fast Ethernet, Gigabit Ethernet<br>Fast Ethernet Gigabit Ethernet. |  |

| | | | | |
|---|---|---|---|---|
| STP | category 6, 6a | 10 Gbps | advantages:<br>* Shielded.<br>* faster than UTP.<br>* Less susceptible to noise and interference. | |
| SSTP | Category 7 | 10 Gbps | disadvan:<br>* expensive<br>* Greater installation effort. | |
| Coaxial cable | RG-6<br>RG-59<br>RG-11 | 10 - 100 Mbps | Adv:<br>* High bandwidth<br>* Immune to interference.<br>* Lowless bandwidth<br>disadv:<br>* Limited distance<br>* Cost.<br>* Size is bulky. |  |
| Fibre optic Cable | Single Mode<br>multimode | 100 Gbps | Adv:<br>* High speed<br>* High bandwidth<br>* High security.<br>* Long distance.<br>dis:<br>* expensive<br>* Requires skills Installer. | |

Result: Thus, we studied the diff types of network cables.

19.5/21